Immersive Exploration of Air Traffic Networks: A Literature Review of Network, Geographic, Temporal and Voice-Based Immersive Analytics

Alper Erdoğan
Polytech Angers – Projet 5A SAGI "Data Exploration"
Supervisor: Paul Richard

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Abstract

[TODO: 150–250 words. Context (immersive analytics of networks), the gap (geographic + temporal networks with real-time voice interaction), method (number of papers, sources, years), main findings per theme, research gaps that motivate FlightlyVR.]

Keywords: immersive analytics; network visualization; virtual reality; flow maps; air transport networks; natural language interfaces; large language models.

1 Introduction

Immersive analytics studies how virtual and augmented reality can support data understanding and decision making [10, 26]. Node-link networks are one of its most studied data types [18]. [TODO: Motivation: why air traffic networks, why VR, why voice. One paragraph each.]

Scope. This review covers work at the intersection of (i) immersive network visualization, (ii) geographic flow visualization in VR, (iii) dynamic (time-varying) networks, and (iv) speech and natural language interaction for visualization, with air transport networks as the application domain.

Research questions.

RQ1 Which layouts (3D, egocentric, geographic) have been evaluated for networks in VR, and with which results?

RQ2 How have geographic flows (origin–destination data) been shown in immersive environments?

RQ3 How is time represented and navigated in immersive network visualization?

RQ4 How have speech, natural language and LLM-based assistants been used to interact with (immersive) visualizations, and how were they evaluated?

RQ5 Which gaps remain for geographic, temporal networks explored with real-time voice?

Contributions. [TODO: Taxonomy / comparison table, identified gaps, positioning of FlightlyVR.]

Related surveys. Earlier surveys cover immersive analytics in general [12], abstract 3D visualizations [22], open challenges [10] and immersive network analysis [18] (Section 3). [TODO: What this review adds over them (domain: air transport; voice/LLM; time).]

2 Review Method

The review follows the guidelines of Kitchenham and Charters [20] and reports the selection with a PRISMA 2020 flow diagram [27].

2.1 Sources and search strings

Source	Search string / note
IEEE Xplore	`("virtual reality" OR immersive OR "head-mounted") AND (graph OR network OR "node-link")`
ACM DL	same
Eurographics DL	same (EuroVis, PacificVis, CGF)
Scopus	same, plus `AND (flight OR "air transport" OR aviation)`
arXiv (cs.HC, cs.GR)	preprints from 2024 on (fast-moving LLM / voice topics)
Snowballing	backward and forward from [18, 22]

Table 1: Sources and search strings.
[TODO: Date of each search.]

2.2 Inclusion and exclusion criteria

I1 Peer-reviewed paper or preprint, in English, 2005–2026.

I2 Presents or evaluates a visualization of network, flow or geographic data in VR/AR, or speech / natural language interaction with a visualization.

E1 Desktop-only work without a transferable finding (kept only as background).

E2 Medical / molecular 3D rendering without network structure.

E3 Short abstracts without results; duplicates (keep the longest version).

2.3 Selection

2.4 Coding scheme

Each included paper is coded on: data type (abstract network / geographic / dynamic), layout (2D, 3D, egocentric, map, globe), display (HMD, CAVE, AR), interaction (controller, gesture, gaze, speech, LLM), tasks (following the graph task taxonomy of Lee et al. [24]), evaluation (none, case study, controlled study; N; measures such as SUS [6], NASA-TLX [15], SSQ [19]) and main finding. [TODO: Who coded, how disagreements were resolved.]

3 Surveys and Research Agendas

Marriott et al. [26] (2018). *Problem.* Immersive analytics was emerging as a field without a shared definition, vocabulary or research agenda. *Method.* Edited book with chapters by researchers from visualization, VR/AR and HCI, each covering one aspect (definitions, interaction, collaboration, evaluation, applications). *Results.* Defines immersive analytics as the use of engaging, embodied analysis tools to support data understanding and decision making, and sets

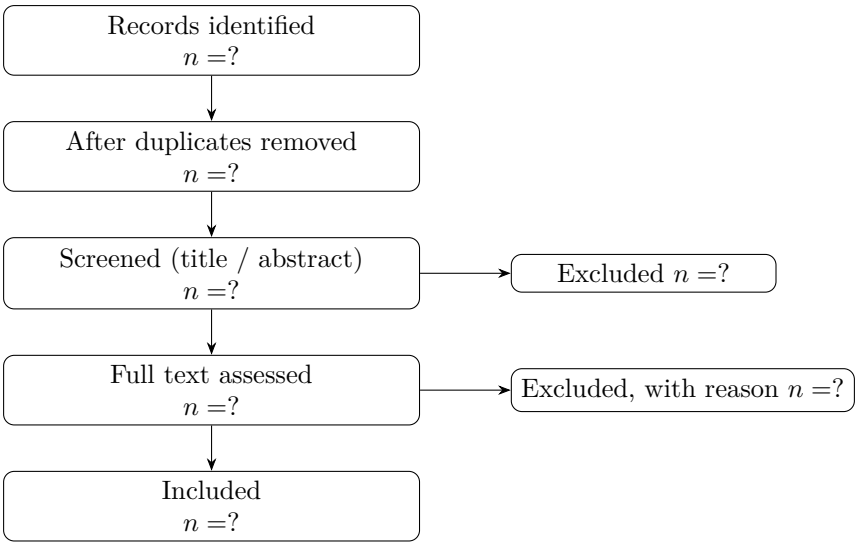


Figure 1: PRISMA-style selection flow.
[TODO: Fill in the counts.]

out the open questions that later work builds on. [TODO: Check the definition wording against the book's first chapter.]

Fonnet and Prié [12] (2021). *Problem.* Immersive data analysis systems had been built for three decades, but no survey covered them. *Method.* Survey of immersive analytics systems from the early 1990s to 2019, classified by rendering technology, data, sensory mapping, interaction and evaluation. *Results.* Multi-sensory aspects are under-used, knowledge from the 3D user interface and VR communities is not applied enough, the field should converge on best practices and aim for real-life systems.

Kraus et al. [22] (2022). *Problem.* Publications on abstract (non-spatial) data in immersive environments had grown quickly but were hard to compare. *Method.* EuroVis state-of-the-art report: a characterization along the use of 3D, embedding in the immersive space, combination with spatial data, interaction paradigms and evaluation, applied to the published works. *Results.* Overview of the state of the art, current trends, and gaps and challenges; the characterization serves as a frame to position new systems.

Ens et al. [10] (2021). *Problem.* A fast-moving field spread over many venues needed a common research agenda. *Method.* 24 international experts, starting from a virtual workshop at ACM CHI 2020, developed challenges over several sessions. *Results.* 17 grand challenges forming a roadmap, among them natural and multimodal interaction and evaluation with real data and users.

Joos et al. [18] (2025). *Problem.* Immersive network visualization is fragmented: the design space is not described and it is unclear which design questions are backed by evidence. *Method.* Structured survey of 138 publications: design-space dimensions, their coverage in 87 applications, and 59 user evaluations; online interactive resource. *Results.* A design space for immersive network analysis, a synthesis of experimental findings, key achievements and research gaps, and design guidance. It is the backbone for Section 4.

4 Immersive Network Visualization

4.1 Layouts: 3D, spherical and egocentric

Kwon et al. [23] (2016). *Problem.* Information visualization had kept to 2D; layout, rendering and interaction methods for graphs in head-mounted displays were barely explored. *Method.* Design of a spherical layout with the viewer at its centre, rendering (including edge bundling and depth cues) and interaction techniques for HMDs; user study against a traditional 2D graph visualization. *Results.* Participants answered significantly faster and with fewer interactions in the immersive condition, especially on harder tasks; overall correctness did not differ, but was significantly higher for larger graphs.

Cordeil et al. [8] (2017). *Problem.* Collaborative analysis of abstract data in CAVE-style rooms versus low-cost HMDs had never been studied formally. *Method.* First user study of collaborative network connectivity tasks in pairs, HMD versus the CAVE2. *Results.* HMD pairs were faster; the CAVE2 led to asymmetric movement between collaborators; accuracy and communication did not differ.

Kotlarek et al. [21] (2020). *Problem.* It was unclear whether an immersive view helps or hurts the mental map users build of a network. *Method.* Comparison of a 3D immersive and a 2D desktop environment on three mental-map tasks: interpreting structure, memorizing nodes and identifying structural changes. *Results.* Structure was interpreted more accurately in the immersive environment, especially for larger networks; the 2D desktop was better for tasks that need spatial memory.

Sorger et al. [30] (2021). *Problem.* Visual clutter and cybersickness limit exploration inside 3D networks. *Method.* "Egocentrism" metaphor: depiction and interaction adapted to the user's position in the network; interfaces with different degrees of egocentrism compared in a user study (visual search, traversal efficiency, spatial orientation, cybersickness). *Results.* A simple egocentric interface clearly improved visual search and navigation without reducing spatial orientation or increasing cybersickness; an occlusion-free "Ego-Bubble" added little. Released as an online tool.

Bauer et al. [2] (2024). *Problem.* Immersive network systems use one static layout, but no layout suits every task. *Method.* Multi-layout system for hierarchical network data in VR that switches layouts and interactions with the task and level of detail; user study against a state-of-the-art immersive baseline at different spatial scopes. *Results.* The multi-layout approach outperformed the baseline both in spatially focused tasks and when the whole network had to be considered.

Takahira et al. [32] (2026). *Problem.* Egocentric networks (an ego and its neighbours) suffer from clutter on the desktop; layouts tailored to immersive space were missing. *Method.* Design properties and dimensions for immersive egocentric layouts; four layouts (Cube, Cylindrical, Radial, Spherical); user study with 24 participants on egocentric analysis tasks. *Results.* Cube worked well for ego–alter connection strength; Spherical was more effective for understanding the neighbours' topology, reducing occlusion and using 3D space.

4.2 Visual encoding of edges

Büschel et al. [7] (2019). *Problem.* There was little knowledge on how to draw edges in 3D node-link diagrams for efficient immersive analysis. *Method.* Exploration of the design space of edge styles in augmented reality and a user study comparing six edge variants. *Results.* [TODO: Add which edge styles performed best (the abstract does not say; read the results section).]

4.3 Navigation and embodiment

Huang et al. [16] (2023). *Problem.* Navigating to different parts of a network quickly is essential, but the effect of different levels of embodied interaction had not been studied. *Method.* Controlled study with 20 participants and four environments, from a 2D desktop with a mouse to 3D VR, on three navigation-heavy network tasks; accuracy, speed, workload and preference measured. *Results.* On the task that required scanning the whole network, VR and a trackball mouse were more accurate than the desktop; VR was the least mentally demanding and frustrating in two of three tasks and the most preferred; comparing two side-by-side networks was hard in VR.

Bozgeyikli et al. [5] (2016). *Problem.* Locomotion strongly shapes the VR experience, and continuous movement can cause motion sickness. *Method.* "Point & Teleport" (point where to be, get teleported) compared with walk-in-place and joystick; a second experiment added choosing the orientation; 16 users. *Results.* Point & Teleport was fun and easy to use and has no visible translation (so it is not expected to cause motion sickness); the added direction component degraded the experience.

Zhang et al. [35] (2025). *Problem.* How users look at networks in VR is unknown. *Method.* Eye-tracking study in VR with 18 participants relating structural properties of the graph to gaze behaviour. *Results.* [TODO: Verify authors, method and results from the paper; the numbers come from the earlier report.]

5 Geographic Flows and Air Traffic

Yang et al. [33] (2018). *Problem.* How to render world-wide geographic maps in VR was open. *Method.* Comparison of four designs: exocentric globe, flat map, egocentric globe (viewer inside) and curved map around the user; the geographic centre can be moved with a controller and users can walk around; tasks on distance, area and direction. *Results.* The exocentric globe was most accurate for distance and most accurate and fastest for direction, and weakly preferred; area took longer on globes than on flat and curved maps; the curved map generally beat the flat map; the egocentric globe was the least effective.

Yang et al. [34] (2019). *Problem.* When displays are no longer flat, how should origin–destination flows be drawn in the space around the user? *Method.* Three studies: 2D and 3D encodings of flows on flat maps, then flat maps, 3D globes and "MapsLink" (a pair of linked flat maps). *Results.* On flat maps, raised flows with height proportional to distance were most accurate and straight 2D lines fastest; MapsLink was slowest; the 3D globe with raised flows was the fastest, most accurate and most preferred. Careful use of the third dimension reduces clutter in flow maps.

Hurter et al. [17] (2019). *Problem.* Large sets of 3D trajectories (aircraft routes, brain fibres) overlap and cross, so individual ones cannot be followed in dense areas. *Method.* FiberClay renders many trajectories in real time on the GPU in VR and adds interaction techniques that use the controllers and the user's position to compose 3D queries. *Results.* Analysts can select and compare sets of trajectories by geometry and data properties; insights are reported with domain experts in air traffic control and neurology.

Guimerà et al. [14] (2005). *Problem.* The global structure of the air transportation network and the roles of its cities were not understood. *Method.* Network analysis of the worldwide air transportation network: degree, centrality, community detection and each city's pattern of inter-

and intra-community links. *Results.* The network is scale-free and small-world, but the most connected cities are not always the most central; this comes from its multi-community structure, which geography alone does not explain (geopolitics matters); each city gets a global role.

6 Time-Varying Networks

Sorger et al. [29] (2019). *Problem.* Large dynamic networks are analysed in 2D, and immersive solutions scale poorly and rely on fly- or walk-through navigation. *Method.* VR system that interweaves overview exploration and immersive detail analysis, on a web-based 3D engine supporting several input modalities. *Results.* Validated with a performance evaluation and an expert case study on a co-morbidity network; no controlled user study.

Filipov et al. [11] (2023). *Problem.* Structural (node-link vs. matrix) and temporal (animation vs. juxtaposition) encodings of dynamic graphs had rarely been compared together. *Method.* Study of response time, accuracy and preference for combinations of structural and temporal encodings on typical dynamic network tasks, with and without interaction. *Results.* Matrices supported tasks on low-level entities better; interaction made answers slower but more accurate; node-link with automatic animation was fastest and most accurate overall; animation with playback control was the preferred temporal encoding.

7 Speech, Natural Language and LLM Interaction

Bolt [4] (1980). *Problem.* Graphics interfaces relied on keyboards and devices; users could not simply say and show what they wanted. *Method.* "Put-that-there" in MIT's Media Room: speech recognition combined with pointing at a large wall display to create, move and change objects. *Results.* Deictic words ("that", "there") are resolved by pointing, showing that voice and gesture together make a natural interface; a demonstration system, without a formal user study.

Srinivasan and Stasko [31] (2018). *Problem.* Visualization tools are mostly WIMP-based; natural language and direct manipulation were rarely combined, and network visualization had almost no natural language interaction. *Method.* Architecture and prototype (Orko) for network visualization accepting both natural language and touch-based direct manipulation; initial user evaluation. *Results.* The two modalities complement each other; observations give opportunities and challenges for multimodal network visualization interfaces.

Badam et al. [1] (2017). *Problem.* Which input modalities suit visual data exploration in immersive environments. *Method.* Workshop paper analysing the affordances of input modalities (such as touch, speech, gesture and gaze) for immersive visual exploration. *Results.* [TODO: Add the main recommendations from the paper.]

Song et al. [28] (2025). *Problem.* How users phrase spoken requests in immersive analytics, and how they use embodiment cues ("this", "here") in speech, was underexplored. *Method.* Wizard-of-Oz study with 15 participants; axial coding of 1,280 speech acts from 734 utterances; speech input uncertainty measured with semantic entropy. *Results.* Five speech input patterns: users blend embodied and non-embodied speech depending on task, phase and reliance on embodiment cues; embodiment cues raise uncertainty for the system; design implications per pattern.

De La Torre et al. [9] (2024). *Problem.* Creating and changing interactive mixed reality worlds in real time with LLMs fails when training data is scarce or behaviour must be synthesized. *Method.* LLMR framework on Unity: scene understanding, task planning, self-debugging and

memory management around GPT-4; evaluation on creation and modification tasks; usability study ($N = 11$). *Results.* About four times lower average error rate than plain GPT-4; diverse objects, tools and scenes produced and edited; positive participant experiences.

Lee et al. [25] (2026). *Problem.* Authoring operations of visual network analysis lack UI affordances in immersive environments, and multimodal input has a steep learning curve. *Method.* Research-through-design study: a system with LLM-supported voice interaction for immersive network visualization; user study on social network analysis with participants from social and computer science. *Results.* Voice raised perceived usability over controllers and lowered the effort of formulating commands, as users stated intent in natural language; it supports complex, multi-parameter operations. Closest work to FlightlyVR.

8 Methods and Instruments Used

Fruchterman and Reingold [13] (1991). *Problem.* Drawing undirected graphs with straight edges in a readable way. *Method.* Modification of Eades' spring embedder: forces modelled on natural systems, aiming at uniform edge lengths. *Results.* A simple, intuitive and efficient force-directed algorithm; used by FlightlyVR for the 3D shell layout.

Blondel et al. [3] (2008). *Problem.* Community detection methods were too slow for very large networks. *Method.* Heuristic modularity optimization (the "Louvain" method). *Results.* Faster than all other known methods with high modularity; shown on a 2-million-customer phone network and a 118-million-node web graph; used by FlightlyVR for regional clusters.

Lee et al. [24] (2006). *Problem.* Designers and evaluators of graph visualizations lacked a detailed, shared task list. *Method.* Definition of graph-specific objects and a list of commonly encountered tasks. *Results.* Complex tasks can be described as series of low-level tasks on those objects; the taxonomy supports comparable controlled experiments. Basis of the study tasks.

Brooke [6] (1996). *Problem.* Industry needed a quick, reliable usability measure. *Method.* System Usability Scale: ten statements on a five-point agreement scale. *Results.* A single 0–100 usability score, now the most widely used usability questionnaire.

Hart and Staveland [15] (1988). *Problem.* Measuring the mental workload of operators. *Method.* Development of the NASA Task Load Index from empirical and theoretical research: six subscales (mental, physical and temporal demand, performance, effort, frustration) with pairwise weighting. *Results.* A multidimensional workload score, standard in HCI and VR studies.

Kennedy et al. [19] (1993). *Problem.* The motion sickness questionnaire in use did not fit simulator sickness. *Method.* Factor analyses of more than 1,100 questionnaires from 10 Navy simulators, validated on 3,691 simulator flights. *Results.* The Simulator Sickness Questionnaire with simple scoring and better diagnostic power; used to measure cybersickness in VR.

Kitchenham and Charters [20] (2007). *Problem.* Literature reviews in computing were often unsystematic and not reproducible. *Method.* Guidelines adapted from medical research for planning, conducting and reporting systematic reviews. *Results.* A protocol (research questions, search strategy, selection, quality assessment, data extraction) followed in Section 2.

Page et al. [27] (2021). *Problem.* The 2009 PRISMA statement was outdated by advances in review methods. *Method.* Updated guideline built by an expert group. *Results.* A 27-item checklist, an abstract checklist and new flow diagrams; used for Figure 1.

9 Comparison

Work	Network	Geographic	Time	Voice/NL	LLM	User study
Kwon et al. [23]	✓	–	–	–	–	✓
Sorger et al. [29]	✓	–	✓	–	–	○
Yang et al. [34]	○	✓	–	–	–	✓
Hurter et al. [17]	–	✓	○	–	–	○
Bauer et al. [2]	✓	–	–	–	–	✓
Srinivasan, Stasko [31]	✓	–	–	✓	–	✓
Lee et al. [25]	✓	–	–	✓	✓	✓
FlightlyVR	✓	✓	✓	✓	✓	?

Table 2: Features of the closest works (✓ yes, ○ partly, – no).
[TODO: Check every cell against the papers; add rows from the coding.]

10 Research Gaps and Positioning

G1 Geographic and topological layouts of the same network have been studied separately ([23] vs. [33, 34]); switching between them in one immersive system is little explored.

G2 No controlled study compares a VR time slider for dynamic networks with an alternative ([29] is a case study).

G3 Voice interaction for immersive networks uses speech-to-text pipelines on abstract social networks [25]; real-time speech-to-speech models that call application tools, on geographic and temporal data, are not studied.

G4 Evaluations rarely use real, large operational datasets [10].

[TODO: How FlightlyVR addresses each gap; planned user study.]

11 Threats to Validity

[TODO: Search strings may miss work; preprints (arXiv) are not peer reviewed; single coder; fast-moving LLM topic (cut-off date of the search).]

12 Conclusion

[TODO: Answer RQ1–RQ5 in one paragraph each or one short list.]

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